



MERIDIAN JUNIOR COLLEGE
JC2 Preliminary Examinations 2017
Higher 2

CANDIDATE
NAME

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CIVICS
GROUP

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INDEX
NUMBER

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H2 BIOLOGY

9744/04

Paper 4 Practical

12 September 2017

2 hours 30 minutes

Candidates answer on the Question Paper.

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write your name, civics group and index number on all the work you hand in.

Give details of the practical shift and laboratory, where appropriate, in the boxes provided.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams and graphs.

Do not use staples, paper clips, highlighters, glue or correction fluid/tape.

Answer **all** questions in the spaces provided on the Question Paper.

The use of scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

Shift
Laboratory

For examiner's Use	
1	/ 21
2	/ 20
3	/ 14
Total	/ 55

This paper consists of **19** printed pages.

[Turn over]

Answer **all** questions.

QUESTION 1

Fig. 1.1 shows the structure of a dormant seed that has been cut in half (longitudinal section).

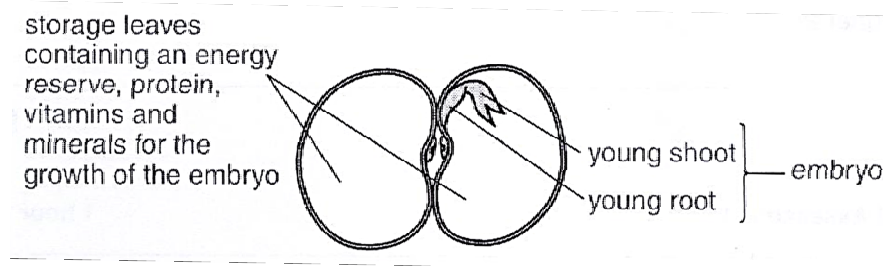


Fig. 1.1

Dormant seeds have a very low rate of respiration. When water is absorbed by dormant seeds, growth hormones are activated. These hormones activate genes that code for the synthesis of enzymes. These enzymes are used to hydrolyze the food reserves so they can be used for respiration and growth. The respiration rate can be measured using a respirometer.

Fig. 1.2 shows the respirometer.

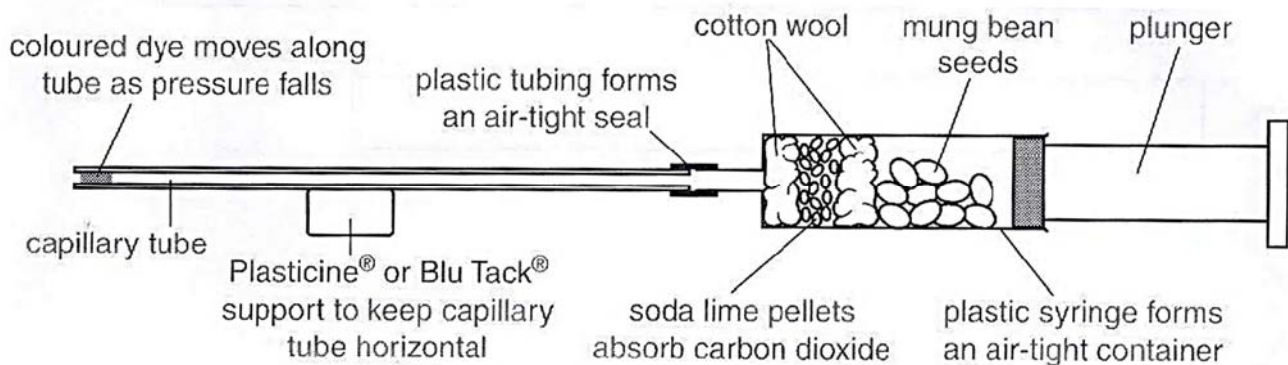


Fig. 1.2

As the seeds respire, oxygen is removed from the air and carbon dioxide is released. This carbon dioxide is absorbed by the soda lime. As the oxygen is used by the seeds, the pressure falls, causing the coloured dye to move along the capillary tube.

You are required to investigate the rate of oxygen uptake by respiring seeds.

Soda lime is harmful and corrosive. Safety glasses and gloves should be worn.

1. Sketch a fully-labelled graph to show the expected relationship between the volume of oxygen uptake and time. [1]

Proceed as follows:

Fig. 1.2 shows the respirometer that you will be setting up and using.

2. Place a small plug of cotton wool in the respirometer and use a glass rod to gently push it to the bottom. The cotton wool must **NOT** be compacted.
3. Using a spatula, add soda lime pellets on top of the cotton wool plug in the respirometer. The soda lime pellets should form a layer of about 1 cm deep.
4. Add another small plug of cotton wool to the respirometer and gently push it down until it is just above the soda lime pellets. Do **NOT** compact the cotton wool.
5. Take 10 mung beans and briefly dab them dry with paper towels. Find and record their mass.

Mass: g

6. Add the 10 mung bean seeds to the respirometer. Place the plunger. Leave the respirometer for 5 minutes.

Explain the purpose of leaving the respirometer for 5 minutes. [1]

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7. Read through the remaining steps up to step **15** and decide on the results that you will be recording. Prepare a table to record all of these results in the space provided under step **15**.
8. Place the end of the capillary tube into the dye solution and use the syringe plunger to pull in about 1 cm length of the dye. Wipe off any excess dye on the outside of the capillary tube.
9. Place a ruler alongside the capillary tube and use a support, such as Plasticine® or Blu Tack®, to keep the capillary tube horizontal and aligned with the ruler.
10. Record the **cumulative distance** moved by the dye at every **30-second intervals** for **5 minutes**. Do not start the stopwatch until the dye has started to move.

- 11.** If the dye has reached the end of the capillary tube before 5 minutes, pause the stopwatch but do not reset it to zero. Make a note of the cumulative distance moved by the dye up to this point. Reset the respirometer by pushing the plunger to move the coloured dye to the start position. As soon as the dye starts moving again, restart the stopwatch and continue recording the cumulative distance moved by the coloured dye at the completion of each 30-second interval, up to 5 minutes. These measurements will need to include both the cumulative distance noted when the respirometer was reset and the distance moved subsequently.
- 12.** After 5 minutes, carefully expel the coloured dye onto a piece of paper towel by pushing in the plunger.
- 13.** Carefully pull the plunger out of the syringe completely, without disturbing the contents. Replace the plunger and leave for 5 minutes. While waiting, proceed to step **16**.
- 14.** After 5 minutes, repeat steps **8** to **11** to measure and record a second set of results for these seeds.
- 15.** Record all of your results in the table you have prepared below. [5]

16. Suggest an advantage of recording the cumulative distance moved by the coloured dye every 30 seconds, instead of only recording the total distance moved after 5 minutes. [1]

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17. Assuming that a 10 mm length of capillary tubing has a volume of 8.0 mm^3 , calculate the mean rate of oxygen consumption of the mung bean seeds per gram of tissue in $\text{mm}^3\text{s}^{-1}\text{g}^{-1}$ over the entire 5 minutes. [3]

Show **all** the steps in your calculation, including relevant units at each step.

Mean rate of oxygen consumption: $\text{mm}^3\text{s}^{-1}\text{g}^{-1}$

18. Describe a control for this experiment and explain the rationale for this control. [2]

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19. Table 1.1 shows the results from an experiment to measure the rate of oxygen consumption of 15 pea seeds at different temperatures. The experiment was repeated three times for each temperature, and the average rate was calculated.

Table 1.1

Temperature / °C	Average rate of oxygen consumption / mm^3s^{-1}	Average rate of oxygen consumption per gram of tissue / $\text{mm}^3\text{s}^{-1}\text{g}^{-1}$
10	8.30	
15	11.0	
20	15.8	
25	20.0	
30	33.5	

- (a) The mean mass of one pea seed is 50.0 mg.

Complete Table 1.1 by calculating the average rate of oxygen consumption **per gram** of tissue in $\text{mm}^3\text{s}^{-1}\text{g}^{-1}$.

Show your working for the result at 10°C, in the space below. [2]

- (b) State and explain the most important variable that needs to remain constant throughout the experiment. [2]

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- (c) The average rate of oxygen consumption was compared between 10°C and 15°C using Student's t-test. The calculated t-value was determined to be 1.740.

Degrees of freedom	Significance level					
	0.50	0.20	0.10	0.05	0.02	0.01
1	1.000	3.078	6.314	12.706	31.821	63.657
2	0.816	1.886	2.920	4.303	6.965	9.925
3	0.765	1.638	2.353	3.182	4.541	5.841
4	0.741	1.533	2.132	2.776	3.747	4.604
5	0.727	1.476	2.015	2.571	3.365	4.032
6	0.718	1.440	1.943	2.447	3.143	3.707

Using the t-distribution table above, explain what conclusions can be drawn from the calculated t-value. [3]

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- (d) In a further investigation, using the same respirometer at 20°C, the soda lime was removed and the experiment repeated. The dye did not move.

Suggest why the dye did not move. [1]

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[Total: 21]

QUESTION 2

Urea, **U**, reacts with water to form aqueous ammonium carbonate. Aqueous ammonium carbonate produces ammonium ions. These form an alkaline solution which causes red litmus paper to turn blue. The time taken for red litmus paper to turn blue can be used to monitor the progress of the reaction.

K is known to play a role in the above reaction. You are required to investigate the effect of concentration of solution **K** on this reaction.

You are provided with:

- 25 cm³ of 10.0%, **K**, which is an irritant.
- 100 cm³ of distilled water, **W**.
- 25 cm³ of a solution of urea, **U**.
- Red litmus paper, each about 6 cm in length.

It is recommended that you wear safety goggles.

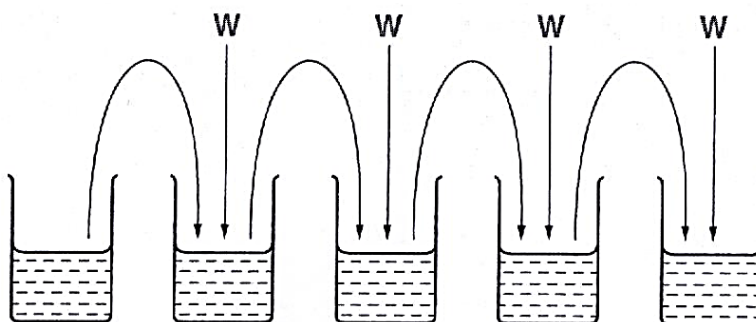
1. Carry out **serial** dilution of solution **K**, to reduce the concentration of the solution by half between each of four successive dilutions, and set up a control.

Label four small beakers, **D1**, **D2**, **D3** and **D4**, for the serial dilutions, and label another small beaker, **C**, for the control.

Complete the table below to show how you will make the different concentrations of solution **K** and how you will set up the control, **C**.

Record the values to **3 significant figures**.

[2]



Label	K	D1	D2	D3	D4
Concentration of K / %					
Volume of solution K taken from the previous dilution / cm ³					
Volume of distilled water, W / cm ³					

Description of the control, **C**:

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2. In order to monitor the progress of the reaction, in step 4, red litmus paper will be added to each mixture of solution **K** and solution **U**, in a test-tube. To prevent the paper from sticking to the wall of the test-tube, you will need to use the glass rod to add it as follows:

Cut a piece of red litmus paper so that it is a little shorter than the circumference of the glass rod. Moisten the paper and stick it to the end of the glass rod as shown in Fig. 2.1. The glass rod can then be lowered into the mixture of **K** and **U**. The red litmus paper will slip off into the mixture and the glass rod can then be removed.

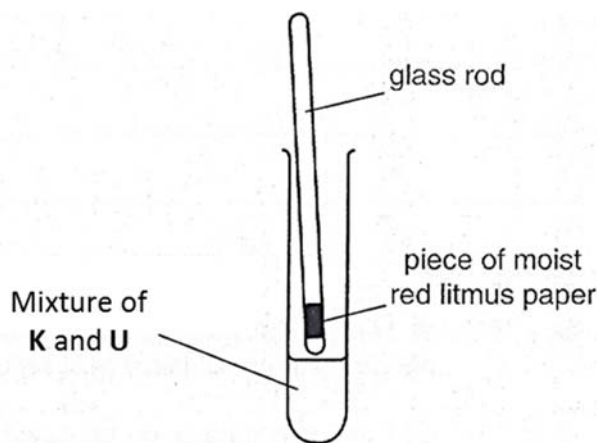


Fig. 2.1

Proceed as follows:

3. To test the activity of the highest concentration of solution **K**, put 2 cm³ of **U** into a test-tube, then add 2 cm³ of **K** and mix well. The reaction will start as soon as **K** is added.

Immediately put one piece of red litmus paper into the test-tube as described in step 2 and start timing.

4. Record the time taken for the piece of red litmus paper to turn blue. If the piece of red litmus paper does not turn blue in 10 minutes, record as 'more than 600'.
5. Repeat steps 3 and 4 for the other concentrations of solution **K**, and the control, **C**. The red litmus paper used each time should be of the same size.
6. Using an appropriate format, record the results of this investigation for the various concentrations of solution **K**, including the control, in the space provided (in step 7).

7. Use the space below to record your results.

[3]

8. From the results of your investigation, suggest the identity of **K**.

[2]

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9. Calculate the rate of reaction using your result for 10.0% concentration of solution **K**. Show your workings clearly.

[1]

Rate of reaction: s⁻¹

10. (a) Lack of replicates is a limitation of this procedure.

Describe one other limitation.

[1]

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(b) Suggest how you would make one improvement to this procedure to reduce the effect of the limitation identified in **10(a)**. [1]

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11. The effect of pH on the activity of two proteolytic enzymes, **A** and **B**, was compared. The substrate of the enzymes was coloured jelly, which is made of proteins. It is known that both enzymes work best at 38°C.

The apparatus for each pH was set up as shown in Fig. 2.2.

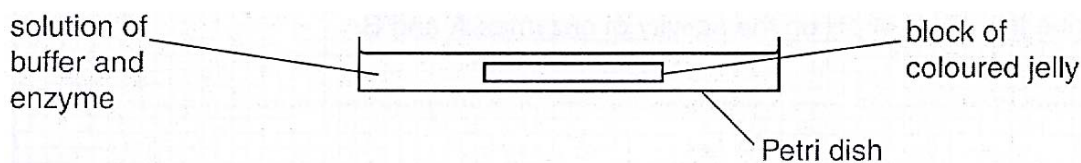


Fig. 2.2

The block of coloured jelly gets smaller as it is digested by the enzymes.

- (a) State two variables which would need to be controlled and suggest how each variable would be controlled. [2]

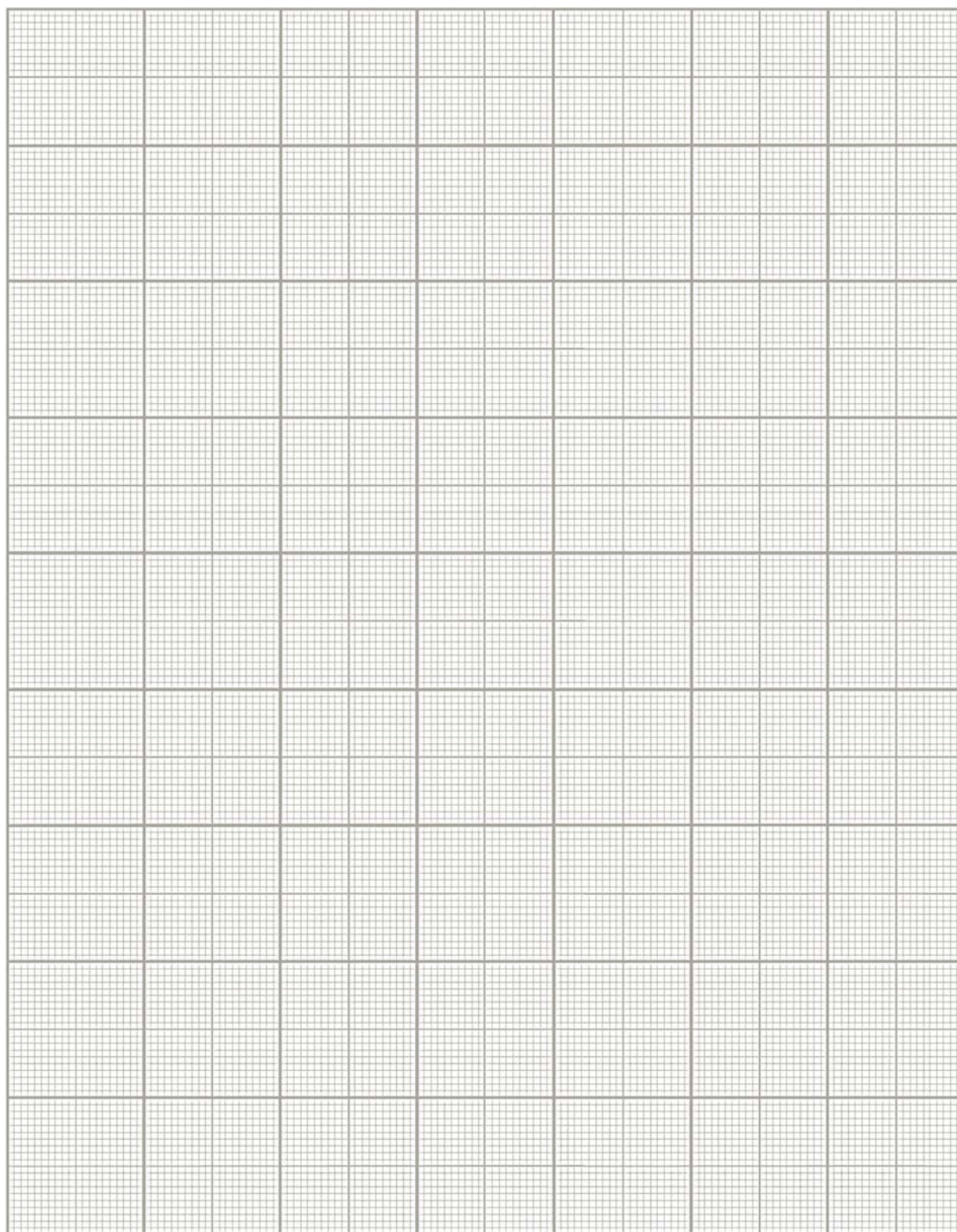
1.
2.

The results of the investigation are shown in Table 2.1.

pH	Area of jelly present after 90 minutes / mm ²	
	enzyme A	enzyme B
4.0	10	134
6.4	76	124
7.4	128	76
8.0	138	52
9.0	140	6

Table 2.1

(b) Plot, on the grid below, the data shown in Table 2.1. Draw lines of best fit for enzyme **A** and enzyme **B**. [3]



(c) Describe the effect of pH on the activity of enzymes **A** and **B**. [1]

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(d) Explain why changes in pH affect the activity of these two enzymes differently. [4]

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[Total: 20]

QUESTION 3

You are required to plan an investigation to find out the effect of surface area-to-volume ratio on the rate of diffusion of hydrochloric acid into agar blocks containing phenolphthalein.

Phenolphthalein is an indicator which appears pink at pH higher than 7, and colourless at pH less than 7.

You must use:

- Sixteen 2cm x 2cm x 2cm phenolphthalein-containing agar blocks at pH 8
- 10 g dm⁻³ hydrochloric acid



You may select from the following apparatus and use appropriate additional apparatus:

- normal laboratory glassware, e.g. test-tubes, boiling tubes, beakers, measuring cylinders, glass rods, etc.
- syringes
- timer, e.g. stopwatch
- white tile
- scalpel
- 15-cm ruler

Your plan should:

- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it.
- be illustrated by relevant diagram(s), if necessary, to show, for example, the arrangement of apparatus used.
- identify the independent and dependent variables.
- describe the method with scientific reasoning used to decide the method so that the results are as accurate and repeatable as possible.
- include the layout of result tables and graphs with clear headings and labels.
- use the correct technical and scientific terms.
- include reference to safety measures to minimise any risks associated with the proposed experiment.

[Total: 14]

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